

Seat No.	
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Set **P**

**B.Sc. (E.C.S.) (Semester - IV) (New) (NEP CBCS) Examination:
March/April – 2026
Statistics for Data Science - II (G06-DSC2-0403)**

Day & Date: Monday, 20-04-2026
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

- Instructions:** 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of Log table and Calculator is allowed.

Q.1 Multiple choice questions. 06

- 1) In _____ number of ways a committee of 5 be selected from 10 persons, such that 2 particular persons are always included and 1 particular person is always excluded from committee.
 - a) $^{10}C_5$
 - b) 8C_3
 - c) 7C_3
 - d) 7C_2
- 2) A random experiment is resulted in to _____ outcomes.
 - a) 1
 - b) 2
 - c) 6
 - d) more than one
- 3) Let 'A' be an event defined on sample space 'S', then $P(\bar{A}/A) = \underline{\hspace{2cm}}$.
 - a) 0
 - b) 1
 - c) $1 - P(A)$
 - d) None of these
- 4) If $X \rightarrow B(10, 0.4)$ then $Var(X) = \underline{\hspace{2cm}}$.
 - a) 4
 - b) 6
 - c) 0.24
 - d) 2.4
- 5) If $X \rightarrow N(5, 100)$ then _____.
 - a) Mean = Median = mode = 5
 - b) $P(X \leq 5) = P(X \geq 5)$
 - c) S.D. = 10
 - d) All of these
- 6) Reject H_0 when H_1 is true is _____.
 - a) Type-I error
 - b) Type-II error
 - c) Correct decision
 - d) None of these

Q.2 Answer the following. (Any Three) 06

- a) Define - parameter and statistic.
- b) State multiplication principle of counting.
- c) If $P(A) = 0.6, P(B/A) = 0.8$ find $P(A \cap B)$.
- d) If ${}^nC_x = {}^{(n+1)}C_{(x+1)}$ comment on values of n and x.
- e) If $X \rightarrow U(10, 12)$ then find $P(X \leq 11)$.

Q.3 Answer the following. (Any Two)**06**

- a) Define - permutations and combinations, state their formulae.
- b) Two identical boxes, Box-I contains 6 white and 5 black balls, Box-II contains 4 white and 3 black balls. A box is selected at random and a ball is drawn from it. Find probability of getting black ball.
- c) A discrete r.v. X has following p.m.f.

X	-1	0	1	2
$P(x)$	$2k$	k	k	$3k$

Find

- i) k
- ii) $P(X < 1)$

Q.4 Answer the following. (Any Two)**06**

- a) State any three properties of normal distribution.
- b) Define - Null hypothesis, Alternative hypothesis and level of significance.
- c) A discrete r.v. X has Poisson distribution with $P(X = 2) = P(X = 3)$.
Find $P(X = 0)$

Q.5 Answer the following. (Any One)**06**

- a) Explain chi-square test for testing independence of attributes.
- b) Marks obtained by students follows normal distribution with mean 55 and s.d. 24. How many students out of 500 have scored marks more than 79.
(Given: area under s.n.v. curve from 0 to 1 is 0.3413)